

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
3rd Year Odd Semester Examination 2023

COURSE NO: CSE 3109

COURSE TITLE: Applied Statistics and Queuing Theory

FULL MARKS: 60

TIME: 3 HRS

N.B. (i) Answer any SIX questions taking any THREE from each section.

(ii) Figures in the right margin indicate full marks.

(iii) Use separate answer script for each section.

SECTION : A

- Q.1. (a) "Statistics is crucial in a wide range of fields, including business, healthcare, engineering, and social sciences" - Justify the statement by mentioning how as an engineer you can use statistics in designing experiments and improving processes. CO1 2
- (b) Determine whether the data below qualifies as a parameter or a statistic. CO1 3
 Clarify your response with a sound justification also.
 (i) The calculated mean household income in a national census where every household is surveyed.
 (ii) The average age of a group of 100 randomly selected people from a city.
 (iii) A manufacturer claims that the average lifetime of a certain type of light bulb is 1,000 hours.
- (c) Let's say a supermarket wants to examine its weekly consumer traffic. Over the course of seven days, they counted the number of consumers who entered the shop at 30-minute intervals between 8:00 AM and 10:00 PM. Here is the data for one day:
 12, 15, 25, 22, 35, 28, 40, 38, 50, 60, 55, 45, 30, 25, 20, 18, 35, 40, 55, 50, 38, 42, 35, 30, 25, 20, 18, 10.
 Now categorize the number of consumers into ranges or intervals and construct a frequency table. Determine the responses to the following queries using the frequency table that you generated:
 (i) The percentage of the intervals having more than 40 customers.
 (ii) The relative and the cumulative relative frequency of the interval at which the peak of consumer traffic occurs.
 (iii) The ogive graph that uses the cumulative relative frequencies of the consumers entered at the shop.
 (iv) A histogram that shows the frequency distribution of the customers entered at the shop.
 (v) The stem-and-leaf display for the entered consumers at the shop on that day.
- Q.2. (a) "The range does not provide any insights into the characteristics of the distribution between the two extreme observations" - Explain with an example. CO2 2
- (b) A study evaluated the rice yield (in kilograms) from two different regions in Bangladesh. The data collected from each region is summarized below: CO2 8
- | Rice yield (kg) | 0-500 | 500-1000 | 1000-1500 | 1500-2000 | 2000-2500 |
|----------------------|-------|----------|-----------|-----------|-----------|
| Frequency (Region A) | 5 | 15 | 30 | 20 | 10 |
| Frequency (Region B) | 2 | 10 | 25 | 30 | 8 |
- Compare the coefficients of variance for the two regions and explain what this indicates about the relative variability of rice yields in each region.
- Q.3. (a) Briefly describe the following terms: CO1 3
 (i) Skewness and Modalities of Histogram
 (ii) Line graph and Frequency polygon
 (iii) Scatter plot and Dot plot
- (b) Professor Yan records the scores of his two classes on a class test. Class A has a standard deviation of 5, while class B has a standard deviation of 15. Based on the class test, what conclusions can you draw about each class's consistency in performance? CO2 2
- (c) The following data table shows the height of the plants after 4 weeks (in cm): CO2 5
- | Height Range | 0-5 | 6-10 | 11-15 | 16-20 | 21-25 |
|--------------|-----|------|-------|-------|-------|
| Frequency | 2 | 6 | 9 | 5 | 3 |
- (i) Determine the following terms from the given data distribution:
 A. Mean, Variance and Standard Deviation of the plant heights
 B. The coefficient of variance (CV) of the plant heights
 C. The mean and quartile deviation of the plant heights
 D. The height ranges that fall within $\pm 2\sigma$ of the mean
- (ii) If another group of plants grown under different conditions resulted in a mean height of 15 cm with a standard deviation of 3 cm, compare the relative growth performance of the two groups based on your quantitative analysis.
- Q.4. (a) A group of fishermen in Bangladesh goes fishing for Hilsa fish on two consecutive days. The probability of catching Hilsa fish on the first day is 0.7, and the probability of catching Hilsa fish on the second day is 0.5. Assuming that the events of catching fish on both days are mutually independent, determine the probability that the fishermen catch Hilsa fish on both days. CO4 3

- (b) An investor in the Dhaka Stock Exchange has determined that a particular investment strategy has a 70% success rate in generating positive returns. If the investor applies this strategy to 15 different stocks, determine the probability that-

- (i) Exactly 10 of these stocks will yield positive returns.
(ii) At least 1 of these stocks will yield positive returns.

- (c) During the Liberation War of Bangladesh, it was reported that, on average, there were 5 casualties per week in a specific region. Determine the probability that exactly 3 casualties occurred in a given week during the war.

SECTION : B

- Q.5. (a) Prove that the central moment about the mean for ungrouped data is zero. CO2 5

- (b) One card is selected at random from 25 cards numbered 1 to 25. Find the probability that the number on the card is: (i) even, (ii) divisible by 3, (iii) even and divisible by 3, (iv) even or divisible by 3, (v) ends in the digit 2. CO4 5

- Q.6. (a) "Measures of variation tell us about the amount of the variation where, the measures of skewness tell us about the direction of variation" - demonstrate the required figures to support the statement. CO2 (3)

- (b) An analysis of electricity consumption resulted in the following distribution: CO2 4

Consumption (kw/h)	0-10	10-20	20-30	30-40	40-50
No. of Users	6	25	36	20	13

- (c) Calculate the coefficient of skewness and comment on the result.
Assume that a factory named Widget Excellence Ltd. produces widgets, and the quality control department monitors the weights of these widgets. They find that the weights of the widgets have a high kurtosis. Analyze the implications of the high kurtosis on the production process of the factory, indicating the consistency of widget weights. CO2 3

- Q.7. (a) An electronic device has two components, C1 and C2, but it will operate if at least one of the components is working properly. Each component has its own switch, and both switches must be turned on, one after the other, for the device to begin operating. Thus, the device can begin to operate by either switching C1 on first and then C2, or vice-versa. If C1 is switched on first, it fails immediately with probability 0.01, whereas if C2 is switched on first, it fails immediately with probability 0.02. Furthermore, if C1 is switched on first and fails, the probability that C2 fails immediately when it is switched on is 0.025, due to added strain. Similarly, if C2 is switched on first and fails, the probability that C1 fails immediately when it is switched on is 0.015. Determine which of the two switches (C1 or C2) should be turned on first to ensure optimal device operation. CO4 (3)

- (b) Suppose, you are a member of a class of 18 students. A bowl contains 18 chips: 1 blue and 17 red. Each student is to take 1 chip from the bowl without replacement. The student who draws the blue chip is guaranteed a 100% marks for the course. If you have a choice of drawing first, fifth or last, which position would you choose? Justify your choice on the basis of probability. CO4 3

- (c) Suppose a random variable X takes on the values -3, -1, 2, and 5 with respective probabilities:

$$\frac{2K-3}{10}, \frac{K-2}{10}, \frac{K-1}{10}, \frac{K+1}{10}$$

- (i) Determine the distribution table of X.

- (ii) Find the expected value E(X) of X.

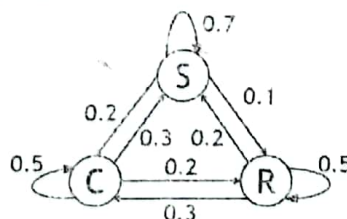
- (d) Let X be a continuous random variable with the cumulative distribution function (CDF) given by: CO4 2

$$F_x = \begin{cases} 0, & x < 0 \\ x/4, & 0 \leq x \leq 4 \\ 1, & x > 4 \end{cases}$$

Calculate $P(1 < x \leq 3)$.

- Q.8. (a) Describe the M/M/1 queuing model briefly. CO5 4

- (b) A weather forecasting model predicts the weather conditions in a city as either Sunny (State S), Cloudy (State C), or Rainy (State R). This is modeled using a Markov Chain in the figure below. CO5 6



$C \rightarrow C \rightarrow C \rightarrow R$
 $C \rightarrow C \rightarrow S \rightarrow R$
 $C \rightarrow C \rightarrow R \rightarrow R$

- (i) Construct the transition matrix for this Markov Chain.
(ii) If the current weather is cloudy (State C), determine the probability that it will be Sunny (State S) after two days.

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

SECTION : A

COs Marks

- Q.1. (a) A class test with the same questions has been conducted in two sections B and C. Two samples from each sections are taken and their marks have been analyzed. The marks are given below:

Section B: 18, 17, 19, 10, 12, 20, 18, 15, 10, 12

Section C: 8, 10, 20, 20, 15, 20, 20, 19, 18, 7, 10

Based on the data, make the following analysis:

i) Which section has performed better in the class test in general? **mean jader beshi**ii) Which sections student is less likely to copy from each other in the test? **jader standard deviation beshi**

- (b) The government has initiated a survey to identify household in need of assistance as a part of their subsidy program. The government plans to provide financial support to only the 20% of households with the lowest expenditure. Now we need to determine the maximum expenditure threshold for a household to be eligible for the government subsidy from the given data. The data is in the following table:

Expenditure (\$)	0-100	101-200	201-300	301-400	401-500	501-600
Number of Households	250	400	600	800	700	500

- Q.2. (a) A player has scored the following runs across 10 games of cricket:
78, 50, 52, 61, 20, 14, 82, 140, 62, 48

i) Compute the mean, median, quartiles and quartile deviation.

ii) Check for outliers using IQR.

iii) Remove the detected outliers and compute mean, median, quartiles, and quartile deviation.

iv) Analyze the effect of outlier removal in the mentioned descriptive statistical measures.

Explain the following remark with an example: "Causation always implies correlation, but correlation does not necessarily imply causation".

- (c) A skewed distribution has a coefficient of skewness, $SK_p = -1.47$. Suppose the median and mean of the distribution are 80 and 70 respectively. Find the standard deviation of this distribution.

- Q.3. (a) A random sample of apples were selected from local orchard and their diameter was recorded. The mean diameter was 3.5 inches with a standard deviation of 0.2 inches. What is the minimum percentage of diameters that fall between 3.1 and 3.9 inches?

- (b) Recently a bank named Y released some of the data related to loan. The dataset represents information about individuals including their age, income, education level, Credit-score and the corresponding loan amount they have applied for. Now build a linear/non-linear model based on the data by choosing the dataset's two features named Income and Credit score. The data set is given on the following table. Find out the loan amount if credit-score=700 and income=45000. **2, 21, 280 \$**

Age	Income (in \$)	Education (in years)	Credit Score	Loan Amount (in \$)
35	50,000	12	720	250000
45	75,000	16	800	350000
28	40,000	10	650	200000
52	90,000	18	850	450000
39	60,000	14	720	300000
31	55,000	12	680	280000
48	80,000	15	780	400000
36	65,000	13	700	320000

- (c) Explain how you will judge a regression model's performance.

- Q.4. (a) A venture capital equity is considering investing in one of the two different companies- Alpha and Beta. They have collected some data regarding the 12 months past returns on the shares of both these companies.

Monthly return percentage

Alpha	-1	2	3	4	2	1	-1	-2	-3	-2	0	1
Beta	0	2	5	7	9	2	-2	-5	-7	-9	2	0

Based on knowledge of kurtosis which company would the equity invest in if it prefers investing in the company that is likely to have fewer extremes (outlier) in terms of monthly returns.

- (b) Incomplete marks of 10 students of business school are given below. The two regression

lines (Y on X and X on Y) intersect at the point (37.5, 37.5)

Marks	in	45	36	?	39	34	37	32	44	38	28
Business											
Marks	in	47	38	34	44	32	41	?	32	42	30
Economics											

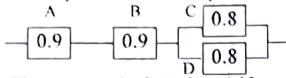
* "?" indicates a missing value

- Adam has scored 40 in Business. Estimate his mark in Economic.
- Smith has scored 40 in Economics. Estimate his mark in Business.

SECTION : B

Q.5. (a) 4 components make up an electrical system as illustrated in the following figure.

CO₁ 4



The system is functional if components A and B work and either of the components C and D works. The reliability (probability of working) of each component is given in the figure. Assume that the 4 components work independently. Find the probability that-

- The entire system is functional.
- Component D does not work given that the entire system is functional.

(b) Bob presents Alice with a box containing 50 coins of which 1 coin is biased with both sides being heads. Alice does not know that a biased coin is inside that box. Bob ask Alice to randomly pick a coin from the box and flip it 4 times. Alice obliges and gets 4 heads. Observing this event, Alice suspects that the coin she selected is biased. Determine the probability of Alice's suspicion being valid.

CO₃ 5

(c) Derive the mean and variance of Bernoulli distribution.

CO₄ 3

Q.6. (a) Samsung produces all its displays at three factories X, Y and Z. Factory X produces 40 percent of the total displays of which 94 percent are non-defective. Factory Y produces 35 percent of the total displays of which 95 percent are non-defective. Factory Z produces 25 percent of total number of displays of which 93 percent are non-defective. If a display is chosen at random and it turns out to be defective, what is the chance that it is produced in factory X?

CO₃ 5

(b) In a data set of 400 emails 80 are labeled as spam and 320 as not spam. Out of the spam emails 70 contain word "click" while out of the non-spam emails, 40 contain the word "click". What is the probability that an email containing the word "click" is spam.

CO₃ 5

(c) What is the main limitation of the naive bayes theorem?

CO₃ 2

Q.7. (a) Suppose a certain type of small data processing firm is so specialized that some have difficulty making a profit in their first year of operation. The probability density function that characterizes the proportion Y that makes a profit is given by-

CO₃ 6

$$f(y) = \begin{cases} ky^4(1-y)^2, & 0 \leq y \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

i) What is the value of k that renders the above a valid density function?

ii) Find the probability that 30 to 80 percent of the firms make a profit in the first year.

(b) An oil drilling company ventures into various locations and its success or failure is independent from one location to another. Suppose the probability of a success at any specific location is 0.20.

CO₄ 6

i) What is the probability that the driller drills at 20 locations and find oil in exactly 2 places.

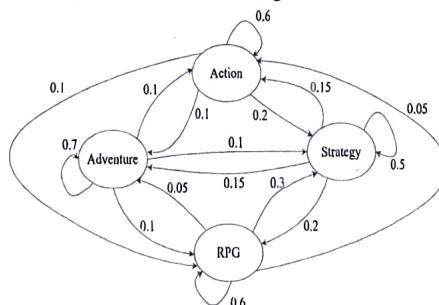
ii) The driller will go bankrupt if it drills 15 times before finding any location with oil. What are the driller's prospects for bankruptcy?

Q.8. (a) Discuss the importance of queuing theory with examples.

CO₅ 4

(b) You are the manager of a small online video games shop that sells games in four genres: Action, Adventure, Strategy and Role-playing games (RPG). Over time you have noticed that the genre of game a customer buys in one month tends to influence what genre they might buy in the next month. This is modeled using a Markov chain in the figure below:

CO₅ 8



- A customer purchased an Action game, then an RPG, then an Adventure game over 3 consecutive months. Find the probability that he will purchase a Strategy game next month.
- Rank the genre according to long term customer purchase behaviour.

*** END ***

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

3rd Year ODD Semester Examination 2021

COURSE NO: CSE 3107 COURSE TITLE: **Applied Statistics and Queuing Theory**

FULL MARKS: 72

TIME: 3 HRS

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

SECTION : A

Marks

- Q.1. (a) Differentiate between descriptive statistics and influential statistics with suitable example. 3
(b) Suppose ABC company produces beverages. The stock of the company is traded both in Dhaka Stock Exchange (DSE) and New York Stock Exchange (NYSE). This week ABC stocks traded at \$63.18, a decrease \$0.74 or 1.16% from the previous trading session. In the past four weeks, it has experienced a 0.16% drop. Over the last year, its price has increased by 11.82%. This company holds 45% share of Bangladeshi beverages market. The following table depicts ABC's performance during past few years on NYSE and DSE. 6

Year	2017	2018	2019	2020
NYSE (%)	3.0 - 4.20	3.2 - 4.60	5.10 - 11.30	2.28 - 3.96
DSE (%)	2.50 - 3.21	4.23 - 6.36	6.53 - 7.50	1.32 - 5.60

- i) Calculate the average yearly rate of return for each stock exchange
ii) Investigate and draw a statistical inference on which stock market we should invest.
- (c) Can we calculate mean for categorical data? Defend your answer with suitable example. 3
- Q.2. (a) Differentiate between statistic and parameter. 3
(b) Consider a scenario where you are interested in finding out batting-bowling average and standard deviation of players of domestic cricket clubs in Bangladesh. There are 500 players in 20 domestic clubs. You randomly select 100 players and calculate their average and standard deviation. What are this study's population, sample statistic, parameter and variables? 4
(c) Elaborate on the following remark: "Temperature expressed in Celsius is interval variable but temperature in Kelvin is ratio variable". 3
(d) Define i) Qualitative variable and ii) Ordinal variable 2
- Q.3. (a) Consider the following runs scored by two batsmen in their previous 10 matches. 3
Batsman V K: 53, 46, 48, 50, 53, 53, 58, 60, 57, 52
Batsman S K: 30, 91, 0, 64, 42, 80, 30, 5, 117, 71
The mean of runs scored by both V K and S K is same i.e. 53. Analyze the performance of two players.
(b) Walmart is one of the major importers of clothing from Bangladesh. It adheres to strict guidelines for selecting clothing items. Consequently, Walmart's rejection rate is higher than any buyer's. The following table depicts a sampling of Walmart's rejections throughout the past month. The minimum number of rejection is 21. 7

#Rejection	Less than 26	Less than 31	Less than 36	Less than 41	Less than 46	Less than 51	Less than 56
#Factory	5	15	28	42	15	12	3

- i) Characterize the nature of distribution of the above tabular data considering the concept of Skewness, Moments and Kurtosis.
ii) Make suitable comments on the above tabular data and draw an appropriate figure to explain the distribution.
- (c) Indicate the significance of Skewness, Moments and Kurtosis with proper examples. 2
- Q.4. (a) The COVID-19 pandemic has a devastating impact on human life. According to statistics, 11 per 1000 COVID-19 test-positive persons dies. Assume that the death toll of COVID-19 test-positive persons follows the Poisson distribution. Of the next 2000 persons who test positive, determine 5
i) The mean number of persons that will die.
ii) The probability that exactly ten (10) persons will die.
(b) Differentiate between mutually exclusive events and independent events with example. Write down the condition for two events to be both mutually exclusive and independent. 3
(c) Assume LK is one of the leading refrigerator manufacturing companies in the world. One of their refrigerator manufacturing plants produces 2000 units per day. The average weight of their refrigerator is 130 kg with a standard deviation of 10 kg. Assuming the weights of refrigerators are normally distributed, how many units are expected to weigh less than 150 kg. 4

SECTION : B

- Q.5. (a) Interpret the values of the Karl Pearson's coefficient of correlation with suitable figure. 3
 (b) The Internet Movie Database (IMDb) is a website that offers information about movies, television shows, home videos and digital streaming contents such as cast, production crew, screenplay, summaries, trivia, ratings and review from critics. A sample IMDb database for several of these movies is presented the table below. 7

#critic_review	723	302	602	813	116	166	234
gross (Normalized)	760	309	200	448	368	191	532
#user_review	3054	1238	994	2701	254	391	205
genres	Action	Action	Thriller	Thriller	Drama	Action	Action
imdb_scores	7.9	7.1	6.8	8.5	6.6	8.3	6.7

Build a linear/nonlinear model to predict the imdb_scores by choosing any of the two features from the above dataset. Finally, predict the imdb_scores for #critic_review=469 and #user_review=2536.

- (c) Briefly explain the "Goodness of Fit in Regression". 2
- Q.6. (a) Historical data shows that the RUET CSE 18 batch has 47 out of 180 students who failed an exam. Assume that past failures affect student's chances of failure in the future examination. The twenty (20) students are randomly selected from RUET CSE 18 and made to attend a mock exam. Determine the probability that in the mock exam- 5

- All 20 students will pass
- At most, 15 students will pass
- Between 4 and 10 students will pass

- (b) Semester final exam mark is assumed to be normally distributed. It is observed that 10% of the students have marks under 65 and 25% exceed 75. What percentages of students have marks between 60 and 74? 5

- (c) Differentiate between empirical probability and theoretical probability. 2

- Q.7. (a) A company 'X' has a policy that among the workers who do overtime, the top 5% in terms of overtime, duration, will get double hourly rate for overtime. The following frequency distribution shows the overtime (measured in hours) done by 100 employees in a month. 5

Overtime	10-15	15-20	20-25	25-30	30-35	35-40
No. of employees	11	20	35	20	8	6

Determine the minimum overtime duration required to get a double hourly rate benefit.

- (b) Random variable X denotes the number of hours a battery can run before it wears out. The probability distribution function of X is: 5

$$f(x) = \begin{cases} Ce^{-\frac{x}{5}}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Find the value of C and calculate the probability that the battery will last more than 10 hours.

- (c) Why Naive Bays classifier is called "Naive"? 2

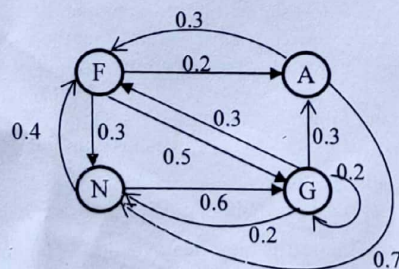
- Q.8. (a) Brain Tumor is a leading cause of death. Brain MRI is the most prevalent method of detecting brain tumor. A computer aided automatic system has been developed using brain MRI images for accurate and precise diagnosis. More than 98% of the cases were correctly predicted by the system. In 5% of instances, it gives false positive results. Brain tumor affects 7% people in developed country on average. 4

Find the probability that:

- No tumor is present given that test is positive.
- Tumor is present given that test is positive.
- Tumor is present given that test is negative.

- (b) List out the fundamental properties of Markov Chain. 2

- (c) PageRank is one the techniques used by Google to determine the importance or relevance of a webpage. In order to implement the PageRank algorithm, we assume that web is a directed graph with web pages serving as nodes and hyperlinks as edges. In the following, directed graph, four (04) nodes designated as F, A, N and G represent four separate web pages. 6



Rank the above pages applying Markov Chain. Assuming that current state is G and there exists at least one stationary state. Iterate five (05) times.

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
3rd Year Odd Semester Examination 2020

COURSE NO: CSE 3107 COURSE TITLE: Applied Statistics and Queuing Theory
 FULL MARKS: 72 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION : A

Marks

- Q.1. (a) What do you mean by joint probability and total probability? Give examples. 3
 (b) There is a 1% probability for a hard drive to crash. Therefore, it has two backups, each having a 2% probability to crash, and all three components are independent of each other. The stored information is lost only in an unfortunate situation when all three devices crash. What is the probability that the information is saved? 3
 (c) Ninety percent of flights depart on time. Eighty percent of flights arrive on time. Seventy-five percent of flights depart on time and arrive on time.
 i) You are meeting a flight that departed on time. What is the probability that it will arrive on time?
 ii) You have met a flight, and it arrived on time. What is the probability that it departed on time?
 iii) Are the events, departing on time and arriving on time, independent? 3
 (d) A speaks truth in 2/5 cases and B lies in 3/7 cases. What is the percentage of cases in which both A and B contradict each other in stating a fact? 3
- Q.2. (a) Explain whether the statement is true or false: The most probable class or label of test example $x = \langle x_1, x_2, \dots, x_n \rangle$ is c_k that maximizes $P(x_1, x_2, \dots, x_n | c_k) P(c_k)$. 4
 (b) According to the recent statistics 7.0% of the population of a certain country has lung disease. Of those having lung disease 90.0% are smokers, of those not-having lung disease 25.3% are smokers. Determine the percentage that a randomly selected smoker has lung disease. 3
 (c) Define mathematical expectation. A lot of 12 television sets include 2 white cords. If 3 of the sets are chosen at random for shipment to a hotel, how many sets with white cords can the shipper expect to send to the hotel? 5
- Q.3. (a) If a variance is binomially distributed, determine its i) mean ii) variance. 3
 (b) The mean grade on a final examination was 72 and the standard deviation was 9. The top 10% of the students are to receive A's. What is the minimum grade that a student must get in order to receive an A? 4
 (c) What is Markov chain? Suppose, in Rajshahi, each day is either sunny or rainy. A sunny day is followed by another sunny day with probability 0.7, whereas a rainy day is followed by a sunny day with probability 0.4. Meteorologists predict an 80% chance of rain on Monday. Make forecast for Tuesday, Wednesday, and Thursday. 5
- Q.4. (a) Explain the Little's Theorem in queuing system with example. 4
 (b) The petrol pump in front of RUET is anxious with its waiting line system and uses a single server system. Based on the previous records, the average number of vehicles arriving per hour is 9 and is described by a Poisson distribution. The service rate is 12 vehicles per hour with the service times following an exponential distribution. The vehicles come from an infinite population. The manager of the pump would like you to calculate the operational characteristics of the system as: (i) the average system utilization (ii) the average number of vehicles in the system (iii) the average number of vehicle waiting in line (iv) the average time a vehicle spends in the system (v) the average time a vehicle spends waiting in line. 5
 (c) Why is queuing model needed in computer science related areas? Explain. 3

SECTION : B

- Q.5. (a) Explain "Correlation does not imply causality". 3
 (b) Explain the following distribution: 4
 i) Exponential distribution
 ii) Normal distribution

- ✓(c) Consider the following data; 5
 20, 28, 29, 30, 36, 37, 39, 42, 53, 54, 55, 58, 61, 67, 68, 70, 74, 81, 82, 93.
 Now, calculate:
 i) Median (M) ii) First Quartile (Q_1) iii) Seventh decile (D_7)
 iv) 25th percentile (P_{25}) v) Inter-quartile range (I_Q)
- Q.6. ✓(a) Define diagnostic statistical analysis. What is the difference between stratified sampling and cluster sampling? 3
 ✓(b) What is the relationship among mean, median and mode for symmetric and differently skewed data? 3
 ✓(c) A car travelled from point P to point Q at 70 kmph and returned at 50 kmph. 4
 i) Which type of mean is more accurate to calculate the mean speed of the whole trip?
 ii) Calculate the mean speed of the whole trip.
- ✓(d) Which technique of measurement of central tendency is most appropriate for data with outliers? 2
- Q.7. (a) What is statistical inference? Explain P-value. 2
 (b) What is matrix of confusion? How to calculate the sensitivity and specificity from the matrix. 3
 (c) The record of GRE quant exam between 1994 and 1997 was 558 ± 139 ($\mu \pm 6$). In a sample of 100 (n) the mean is 585 (M). Compute Z-test for whether or not we will retain the null hypothesis ($\mu = 558$) at a 0.05 level of significance ($\alpha = 0.05$). Also calculate P-value and interpret the results. 7
- Q.8. ✓(a) Illustrate the following plots: 2
 i) Scatter plot ii) Boxplot
 (b) Which attributes are expressed by the following statistical concepts 6
 i) Variance
 ii) Skewness
 iii) Kurtosis
 Explain with appropriate formulas.
- (c) Consider the following table and find 4
 i) Covariance
 ii) Correlation between variables X and Y and interpret results

X	2	8	18	20	28	30
Y	50	45	23	12	5	3
